

ATLIQ TECHNOLOGIES

Enterprise AI Delivery — Forward Deployed Engineering

TECHNICAL DESIGN DOCUMENT

AI-Powered Field Service Command Centre

Prepared for Vaayu Pumps & Systems Ltd

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ABOUT THIS DOCUMENT — READ BEFORE YOU USE IT

This is a TEACHING ARTEFACT from the Codebasics Forward Deployed AI Engineer Roadmap 2026. Vaayu Pumps & Systems Ltd, AtliQ Technologies and every person named in it are fictional.

It is the companion to the Business Requirement Document in this pack. The BRD says what the business needs and why. This document says exactly how the system behaves — precisely enough that a developer who has never spoken to the client, or an AI coding agent, can build it without guessing.

Annex B contains an exercise: feed Section 7 of this document to a coding agent and see whether what comes back matches. That is the real test of whether a spec is finished.

Document Control

Version history

Ver.	Date	Author	Summary of change	Reviewed by
0.1	20 Feb 2026	I. Rao	Initial draft: component design, data model, API specification.	—
0.3	24 Feb 2026	I. Rao	Added severity decision table (Section 7.1) after the rule-set workshop with A. Deshpande. Replaced the earlier prose description, which three reviewers had interpreted differently.	K. Menon
0.5	26 Feb 2026	I. Rao	Integration error codes and idempotency policy added after SAP CPI walkthrough with R. Balan.	R. Balan
1.0	27 Feb 2026	I. Rao	Issued for build. Traceability matrix complete against BRD v1.1 — all 20 functional requirements mapped.	K. Menon

NOTE THE V0.3 ENTRY

The severity rules started life as a paragraph of prose. Three reviewers read that paragraph and each described a different system. Converting it to a decision table (Section 7.1) ended the disagreement in one meeting.

When a rule is important, draw the table. Prose is where ambiguity hides.

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1. Purpose of This Document

This Technical Design Document converts the requirements baselined in **VPS-FSCC-2026-BRD v1.1** into behaviour specific enough to implement and test. It is the last document written before code, and the first document a developer reads.

Document	Answers	Primary audience	Signed by
BRD	What does the business need, and why is it worth doing?	Sponsor, process owner, finance	Business sponsor
FRD	What must the system do, functionally, in business language? (Often folded into the BRD, as it is here.)	Process owner, BA, QA	Process owner
SDD / HLD	What is the shape of the system — components, stack, deployment topology?	Architects, client IT	Client IT / architecture
TDD / LLD	Exactly how does each part behave — contracts, rules, validation, errors?	Developers, QA, the AI coding agent	Technical lead

Commented [SFR1]: Learners confuse these four documents constantly. This table is the answer to the question you will be asked most often. The short version: BRD = what and why, signed by the business. SDD = the shape of the system. TDD = exact behaviour, precise enough to build from. FRD is a middle layer some organisations use and others fold into the BRD or the TDD. Do not fight about the names — agree on who signs what, and move on.

1.1 Scope of this document

In scope: component and agent design, control flow, data model, API contracts, field validation, business rules, integration specifications, model configuration, error handling, security and observability design.

Out of scope: infrastructure provisioning and network topology (covered in VPS-FSCC-2026-SDD), test scripts (covered in VPS-FSCC-2026-UAT), and anything excluded by BRD Section 4.2.

2. Solution Overview

The system is a supervised multi-agent pipeline. A complaint enters on one of three channels, passes through four specialised agents in sequence, and produces a dispatch recommendation that is either auto-approved or routed to a human. Nothing reaches SAP without either passing the confidence gate or being approved by a supervisor.

2.1 Runtime topology

Layer	Contents	Runs on
Ingress	Channel adapters: Graph API mail poller, WhatsApp webhook receiver, coordinator transcript form	On-prem Kubernetes (Vaayu DC, Pune)
Orchestration	LangGraph state machine, four agents, supervisor node, review queue	On-prem Kubernetes
Retrieval	Vector store over manual corpus and 36 months of service history	On-prem (pgvector on the existing PostgreSQL cluster)
Model inference	Primary and fallback LLM endpoints	Azure OpenAI, Central India region (only egress point)
Integration	SAP OData clients, SAP CPI channel for BAPI write, IoT Hub consumer	On-prem Kubernetes
Presentation	Supervisor console (React), KPI dashboard (Grafana)	On-prem Kubernetes
Observability	LangFuse traces, Prometheus metrics, Grafana dashboards	On-prem Kubernetes

Full architecture diagrams, sizing and network design are in the Solution Design Document. This document assumes them.

2.2 Agent topology and responsibilities

Agent	Responsibility	May call	May NOT do
Intake	Normalise the complaint into an intake record; extract structured fields; detect duplicates.	extract_fields, find_open_orders	Assign severity. Contact SAP write APIs.
Triage	Identify equipment; classify severity; diagnose probable fault with citations; determine skill codes and probable parts.	sap_equipment_lookup, retrieve_manual, retrieve_history, sap_contract_lookup	Select a technician. Create anything in SAP.
Dispatch	Rank available technicians; produce a recommendation with a plain-language rationale and a confidence score.	technician_roster, distance_matrix	Approve its own recommendation. Write to SAP.
Parts	Predict probable spare parts; check depot stock; propose alternatives on shortfall.	sap_stock_lookup, depot_distance	Create a stock transfer or reservation.

Agent	Responsibility	May call	May NOT do
Supervisor (node, not LLM)	Apply the routing rules in Section 5.2; hold the review queue; on approval, invoke the SAP write.	sap_order_create	Override the P1 rule. It is not a model and takes no judgement.

TOOL BOUNDARIES ARE THE DESIGN

The right-hand column is doing more work than the rest of the table. In an agentic system, security and correctness come mostly from what each agent cannot reach.

Note that no LLM-driven agent can write to SAP. The only component holding sap_order_create is the supervisor node, which is deterministic code, not a model. That single constraint means a prompt injection in a complaint email cannot create a work order, no matter how clever it is.

3. Data Model

Entities owned by this system are listed below. SAP remains the system of record for equipment, materials, contracts and work orders; the entities here are operational records of this system's own decisions.

Entity	Key attributes	Notes
Complaint	complaint_id (PK), channel, received_at, raw_text, reporter_name, reporter_phone, reporter_email, customer_id, site_id, status	Personal data. Pseudonymised at 24 months per BRD DR-007.
IntakeRecord	intake_id (PK), complaint_id (FK), extracted_fields (jsonb), extraction_confidence, nulled_fields (array), duplicate_of (FK, nullable)	One per complaint. Immutable once written.
TriageResult	triage_id (PK), intake_id (FK), equipment_number, equipment_match_confidence, severity, severity_rule_fired, probable_causes (jsonb with citations), skill_codes (array), sla_tier, sla_due_at	severity_rule_fired stores the rule ID from Section 7.1. Required by BRD FR-018.
PartsAssessment	parts_id (PK), triage_id (FK), predicted_parts (jsonb: material, likelihood band), stock_checked_at, shortfall (bool), alternative_depot	stock_checked_at enforces the 15-minute freshness rule in FR-013.
Recommendation	rec_id (PK), triage_id (FK), parts_id (FK), ranked_technicians (jsonb), selected_technician_id, confidence, rationale_text, routing_reason, status	status: AUTO_APPROVED, PENDING_REVIEW, APPROVED, AMENDED, REJECTED.
Override	override_id (PK), rec_id (FK), supervisor_id, changed_fields (jsonb), reason_code, created_at	Original recommendation is never mutated. BRD FR-012.
DecisionAudit	audit_id (PK), rec_id (FK), agent, model_id, prompt_version, inputs_hash, retrieved_source_ids, output, confidence, latency_ms, token_cost, created_at	24-month retention. BRD FR-018, NFR-08.
TelemetryAlert	alert_id (PK), device_id, equipment_number, metric, value, threshold, breached_at, complaint_id (nullable FK)	Threshold alerts only. No predictive model in Phase 1.

3.1 Retention and personal data

Data	Retention	Rule
Complaint personal fields	24 months	reporter_name, reporter_phone and reporter_email are replaced with a stable pseudonymous hash at 24 months. complaint_id and all linkage survive so audit remains intact.
DecisionAudit	24 months	Retained in full. Contains no personal data by design — inputs are stored as a hash, not as text.
Raw complaint text	24 months	Retained for evaluation. Excluded from any dataset that leaves the Vaayu network.
Telemetry	90 days operational	Older readings remain in Azure IoT Hub archive and are not copied into this system.

4. API Specification

All endpoints are versioned under `/v1` and authenticated with OAuth 2.0 bearer tokens issued by Vaayu's Azure AD tenant. All responses are JSON. All timestamps are ISO 8601 with an explicit offset.

Endpoint	Method	Purpose	Traces to
<code>/v1/complaints</code>	POST	Submit a complaint from any channel adapter	FR-001
<code>/v1/complaints/{id}</code>	GET	Retrieve a complaint with its intake record	FR-001, FR-002
<code>/v1/complaints/{id}/triage</code>	GET	Retrieve the triage result including severity and probable causes	FR-005, FR-006
<code>/v1/queue</code>	GET	Review queue, sorted by SLA time remaining ascending	FR-019
<code>/v1/recommendations/{id}</code>	GET	Retrieve a recommendation with rationale and confidence	FR-011
<code>/v1/recommendations/{id}/approve</code>	POST	Approve; triggers SAP work order creation	FR-017
<code>/v1/recommendations/{id}/override</code>	POST	Amend or reject; reason code mandatory	FR-012, FR-020
<code>/v1/audit/{work_order_id}</code>	GET	Full decision trail for a work order	FR-018
<code>/v1/config/thresholds</code>	GET, PUT	Read and update confidence thresholds; admin role only	FR-009
<code>/healthz</code> , <code>/readyz</code>	GET	Liveness and readiness probes	NFR-02

Commented [SFR2]: An endpoint table is not an API specification. The specification is the next page: one worked contract showing the request, the success response, every error code, and the idempotency rule. Write one of these properly for the most important endpoint, and the pattern for the rest is obvious. Write none, and every developer invents their own error format.

4.1 Worked contract — POST `/v1/complaints`

Request body:

```
{
  "channel": "EMAIL" | "WHATSAPP" | "PHONE_TRANSCRIPT",
  "received_at": "2026-03-04T09:14:22+05:30", // channel receipt, NOT ingestion time
  "raw_text": "Pump at Unit 3 stopped completely this morning...",
  "reporter": { "name": "...", "phone": "...", "email": "..." },
  "customer_ref": "VPS-CUST-04412", // optional; derived if absent
  "attachments": [ { "filename": "...", "content_type": "...", "url": "..." } ],
  "source_message_id": "AAMkAGI2..." // channel-native id, used for idempotency
}
```

Success — 202 Accepted:

```
{
  "complaint_id": "CMP-2026-018842",
  "status": "ACCEPTED",
  "sla_clock_started_at": "2026-03-04T09:14:22+05:30"
}
```

Idempotency

`source_message_id` is the idempotency key. A repeat POST with a `source_message_id` already seen within 7 days returns 200 with the original `complaint_id` and does not create a second record. Channel

adapters retry on network failure, so this is not optional — without it, a WhatsApp webhook redelivery creates a duplicate work order.

Error responses

Code	HTTP	Meaning	Client behaviour
E-INTAKE-001	400	raw_text absent or shorter than 10 characters	Do not retry. Log and alert the coordinator.
E-INTAKE-002	400	received_at missing, malformed, or more than 5 minutes in the future	Do not retry. Adapter clock or mapping fault.
E-INTAKE-003	409	source_message_id already processed	Treat as success. Use the returned complaint_id.
E-INTAKE-004	422	channel not in the permitted enumeration	Do not retry. Configuration fault.
E-INTAKE-005	503	Downstream store unavailable	Retry with exponential backoff: 1s, 2s, 4s, 8s, 16s, then dead-letter queue and alert.
E-INTAKE-006	413	Attachment set exceeds 25 MB	Do not retry. Accept the complaint text without attachments and flag it.

5. Field Validation and Control Flow

5.1 Field-level validation specification

Applies to fields extracted by the Intake and Triage agents. Traces to BRD FR-002 and FR-003.

Field	Mand.	Format and rule	On validation failure
customer_id	Yes	Matches ^VPS-CUST-[0-9]{5}\$ Resolved from reporter email domain or phone against the customer master.	Set null, flag UNRESOLVED_CUSTOMER, route to review queue. Never guess.
equipment_number	Yes	Matches ^[0-9]{12}\$ Fuzzy match permitted against serial, tag number and free description.	If top match confidence < 0.80, return the top 3 candidates and route to review queue. Never auto-select below threshold.
reported_at	Yes	ISO 8601 with offset. Must be within the last 30 days and not in the future.	Fall back to channel received_at and flag REPORTED_TIME_INFERRED.
symptom_text	Yes	Free text, 10 to 4000 characters. Stripped of signature blocks and quoted reply chains before extraction.	If under 10 characters after stripping, reject with E-INTAKE-001.
fault_class	No	One of STOPPAGE, DEGRADED, LEAK, NOISE, ADVISORY. Assigned by the Triage agent.	If confidence < 0.70, leave null. Section 7.1 rule SEV-R9 then applies.
severity	Yes	One of P1, P2, P3, P4. Set only by the rule set in Section 7.1, never by a model directly.	Cannot fail. If no rule matches, SEV-R9 assigns P2 and routes to review.
skill_codes	Yes	Array of codes present in the Vaayu skill matrix. Minimum 1.	Fall back to the baseline skill code for the equipment class and flag SKILL_DEFAULTED.
predicted_parts[].material	No	Matches ^[0-9]{8}\$ and exists in SAP MM material master.	Drop the individual part from the list and log. Do not fail the whole assessment.
reporter_phone	No	E.164. Stored encrypted at rest.	Accept the complaint without it. Not required for triage.

Commented [SFR3]: 'Never guess' in the last column is the single most important design decision in this document. An enterprise AI system that silently invents a plausible equipment number is worse than one that admits it does not know, because the first kind of error is invisible until a technician is standing in front of the wrong machine.

5.2 Routing rules — when a human must decide

Traces to BRD FR-009. The three conditions are evaluated independently; any one of them routes to the review queue. They are deliberately not combined, so that any one can be tuned or disabled without affecting the others.

Rule	Condition	Configurable?	Routing reason recorded
RT-01	severity = P1	No — hard-coded, cannot be disabled	P1_MANDATORY_REVIEW
RT-02	recommendation.confidence < threshold (default 0.85)	Yes — admin, no redeploy	LOW_CONFIDENCE
RT-03	Any upstream integration call failed or returned stale data	No	INTEGRATION_DEGRADED
RT-04	A High-likelihood part is unavailable at the serving depot	No	PARTS_SHORTFALL
RT-05	equipment_match_confidence < 0.80	Yes — admin, no redeploy	EQUIPMENT_AMBIGUOUS

5.3 State machine

A complaint moves through the following states. Transitions not listed are invalid and are rejected by the orchestrator.

State	Entered when	Valid next states
RECEIVED	Complaint accepted at /v1/complaints	EXTRACTED, FAILED_INTAKE, DUPLICATE_LINKED
EXTRACTED	Intake agent completes	TRIAGED, FAILED_TRIAGE
TRIAGED	Triage agent completes; severity assigned	PARTS_CHECKED, FAILED_PARTS
PARTS_CHECKED	Parts agent completes stock check	RECOMMENDED
RECOMMENDED	Dispatch agent produces a ranked recommendation	AUTO_APPROVED, PENDING_REVIEW
PENDING_REVIEW	Any rule in 5.2 fired	APPROVED, AMENDED, REJECTED
AUTO_APPROVED	No rule in 5.2 fired	ORDER_CREATED, ORDER_FAILED
APPROVED / AMENDED	Supervisor action	ORDER_CREATED, ORDER_FAILED
ORDER_CREATED	SAP write succeeded and returned an order number	CLOSED (by SAP, out of this system's control)
ORDER_FAILED	SAP write failed after the full retry policy	PENDING_REVIEW (coordinator alerted; manual IW31 fallback)
REJECTED	Supervisor rejected with a reason code	Terminal

6. Business Rule Specifications

6.1 Severity decision table

Traces to BRD FR-005. Rules are evaluated in order; the first matching rule wins and its ID is written to `TriageResult.severity_rule_fired`.

Rule	Condition	Severity	Notes
SEV-R1	<code>fault_class = STOPPAGE AND equipment.is_critical = true</code>	P1	<code>is_critical</code> is a flag on the SAP equipment master, maintained by Vaayu.
SEV-R2	<code>fault_class = LEAK AND equipment.medium_hazardous = true</code>	P1	Safety overrides commercial tier.
SEV-R3	<code>fault_class = STOPPAGE AND sla_tier IN (PLATINUM, GOLD)</code>	P1	Contractual exposure.
SEV-R4	<code>fault_class = STOPPAGE AND sla_tier IN (SILVER, BRONZE, NONE)</code>	P2	
SEV-R5	<code>fault_class = DEGRADED AND equipment.is_critical = true</code>	P2	Reduced flow, abnormal vibration, overheating.
SEV-R6	<code>fault_class = DEGRADED AND equipment.is_critical = false</code>	P3	
SEV-R7	<code>fault_class = LEAK AND equipment.medium_hazardous = false</code>	P3	
SEV-R8	<code>fault_class = NOISE OR fault_class = ADVISORY</code>	P4	
SEV-R9	<code>fault_class IS NULL (classification confidence below 0.70)</code>	P2	Also sets routing reason <code>LOW_CONFIDENCE</code> . The system never silently downgrades what it does not understand.

Commented [SFR4]: This table is the reason BRD FR-005 is testable. Before it existed, 'assign priority appropriately' meant whatever the reader assumed. Three properties make it work: rules are ordered, first match wins, and the rule ID is stored on the record. That last one is what lets you answer 'why was this a P2?' eighteen months later in an audit.

WHY SEV-R9 ASSIGNS P2 AND NOT P4

When the system does not understand a complaint, it must fail towards attention, not away from it. Defaulting an unclassified complaint to P4 would make the metrics look excellent and quietly bury the hardest cases.

Wherever your design has a default, ask which direction it fails in. Clients rarely ask this. Auditors always do.

6.2 SLA windows and escalation

Traces to BRD FR-015 and FR-016. The clock starts at channel receipt time, not at triage time. Response means first technician contact with the customer; resolution means work order closure.

SLA tier	P1 response	P1 resolution	P2 response	P2 resolution
PLATINUM	2 hours	8 hours	4 hours	24 hours
GOLD	4 hours	24 hours	8 hours	48 hours
SILVER	8 hours	48 hours	16 hours	72 hours
BRONZE	12 hours	72 hours	24 hours	96 hours
NONE (non-AMC)	24 hours (internal target only, no penalty)	Best effort	48 hours	Best effort

Trigger	Condition	Action
ESC-01	Elapsed time reaches 80% of the response window	Warning to the assigned technician and the depot supervisor. Item highlighted amber in the review queue.
ESC-02	Elapsed time reaches 100% of the response window	Escalation to the depot supervisor and the Head of Customer Service. Item highlighted red. Breach recorded against the contract.
ESC-03	Elapsed time reaches 90% of the resolution window	Warning to the depot supervisor with the option to assign a second technician.
ESC-04	Work order is still open at 100% of the resolution window	Escalation to the Head of Customer Service. Penalty flag set for the finance review.

Escalations fire within 60 seconds of the threshold (BRD FR-016). Recipients are configurable per depot. Escalation is evaluated by a scheduled job running every 30 seconds, not by a per-complaint timer, so that a restart cannot lose a pending escalation.

6.3 Technician ranking

Traces to BRD FR-010. Hard filters are applied first; only technicians surviving all filters are scored.

Hard filters (a technician failing any one is excluded)

- Holds at least one of the required skill codes, with certification not expired as at the complaint date.
- On shift, or rostered to start, within the SLA response window for this complaint.
- Road distance from current or home location to the functional location is 150 km or less.
- Not already assigned to an open P1 work order.

Score (applied to survivors; highest score is recommended)

```
score = (0.45 x skill_match)
       + (0.30 x proximity)
       + (0.25 x shift_capacity)
```

```
skill_match  = matched_required_skills / total_required_skills    [0..1]
proximity    = 1 - (road_distance_km / 150)                       [0..1]
shift_capacity = remaining_shift_minutes / 480, capped at 1.0      [0..1]
```

Commented [SFR5]: Weights sum to 1.00 and every input is defined. A developer can implement this without a single clarifying question — which is exactly the bar. Note also that the weights are configuration, not code. The first thing a client asks after go-live is to tune them.

Weights are stored in configuration and are changeable by an administrator without redeployment. Ties are broken by lower current open-order count, then by technician ID ascending, so that ranking is deterministic and reproducible for audit.

6.4 Parts shortfall rule

Traces to BRD FR-013 and FR-014.

Rule	Condition	Action
PRT-01	All High-likelihood parts available at the serving depot	Proceed. Record stock_checked_at.
PRT-02	Any High-likelihood part unavailable at the serving depot	Route to review queue with reason PARTS_SHORTFALL, regardless of confidence. Auto-dispatch is blocked.
PRT-03	PRT-02 fired and an alternative depot within 150 km holds the part	Propose that depot by name, with distance and available quantity.
PRT-04	PRT-02 fired and no depot within 150 km holds the part	State explicitly that no alternative exists within 150 km. Do not propose a partial or speculative option.
PRT-05	Stock reading is older than 15 minutes at the point of recommendation	Re-read before presenting. If the re-read fails, route with reason INTEGRATION_DEGRADED.

7. Integration Specifications

Expands BRD Section 8. One row per interface, plus the shared retry and failure policy below.

ID	Interface	Contract	Failure behaviour
INT-001	SAP PM equipment master (OData API_EQUIPMENT)	GET with \$filter on EquipmentName, SerialNumber or TechnicalObjectDescription. Nightly full sync into the local read model; per-complaintlookup hits the local model first.	Fall back to the local read model with a staleness flag. If the model is more than 36 hours old, route to review with INTEGRATION_DEGRADED.
INT-002	SAP PM order create (BAPI_ALM_ORDER_MAINTAIN via CPI)	POST to the CPI integration flow. Payload carries equipment number, functional location, priority, planner group, assigned technician, skill codes and predicted parts as an operation long text. Idempotency key is rec_id.	Retry 3 times with backoff 2s, 8s, 30s. On final failure, set ORDER_FAILED, alert the coordinator within 60 seconds and present the payload for manual IW31 entry. Never fail silently.
INT-003	SAP MM stock (OData API_MATERIAL_STOCK_SRV)	GET filtered by material and storage location. Response cached 15 minutes per material-location pair.	On failure, route with PARTS_SHORTFALL rather than assuming availability. Failing towards a human is correct here.
INT-004	SAP SD contract (OData API_SERVICE_CONTRACT)	GET by customer. Cached 24 hours.	On failure, apply tier NONE and flag SLA_TIER_UNRESOLVED. Never assume a favourable tier.
INT-005	Microsoft 365 mailbox (Graph API)	Poll every 60 seconds. Messages marked read only after the complaint is durably persisted.	Backoff and continue. Missed poll cycles are recovered on the next successful poll because read-marking is post-persistence.
INT-006	WhatsApp Business Cloud API	Inbound webhook; outbound template messages for technician notification. Signature verified on every inbound call.	Webhook redelivery is expected. Handled by the source_message_id idempotency rule in Section 4.1.
INT-007	Azure IoT Hub telemetry	Event Hub compatible endpoint, dedicated consumer group, checkpointed offsets.	On consumer failure, resume from the last checkpoint. Threshold breaches older than 6 hours are discarded rather than raised late.
INT-008	Equipment manual corpus (SFTP)	Quarterly delta drop. Ingestion pipeline chunks, embeds and versions each document.	Ingestion failure alerts the AtliQ team. The previous corpus version remains live; retrieval is never left empty.

READ THE FAILURE COLUMN AGAIN

Every row fails towards a human or towards the safe assumption. INT-003 assumes the part is missing when it cannot check. INT-004 assumes the least favourable SLA tier when it cannot read the contract.

This is the difference between a demo and a production system. A demo assumes the happy path. A production system decides, in writing and in advance, what it does when each dependency is down — and the client's IT team will ask you exactly this question in the security review.

8. Model and Prompt Configuration

Traces to BRD NFR-09 (cost ceiling of ₹28 per complaint) and NFR-10 (fallback). Model IDs and prompt versions are pinned in configuration and recorded on every DecisionAudit row.

Task	Model class	Temp.	Max out	Rationale
Field extraction (Intake)	Lightweight	0.0	800	Structured extraction against a fixed schema. Deterministic output required.
Duplicate detection	Embedding model	n/a	n/a	Similarity comparison, no generation.
Fault classification	Lightweight	0.0	200	Closed-set classification into five fault classes.
Probable-fault diagnosis	Premium	0.2	1200	The only task requiring synthesis across retrieved manual sections and service history. The single justified use of the expensive model.
Rationale generation	Lightweight	0.3	400	Rewrites already-computed facts into plain language. No reasoning required.
Severity assignment	None — deterministic code	n/a	n/a	Section 6.1 rule table. Deliberately not a model.
Technician ranking	None — deterministic code	n/a	n/a	Section 6.3 scoring formula. Deliberately not a model.

Commented [SFR6]: This table is Principle 6 of the roadmap — cost-efficient AI, not just functional AI — expressed as design rather than intent. Note that the expensive model is used for exactly one task. Most teams reach for the premium model everywhere and discover the unit economics only after go-live, when it is a renegotiation rather than a design choice.

8.1 Fallback chain

1. Primary endpoint: Azure OpenAI, Central India.
2. On 429 or 5xx: retry twice with backoff 1s, 4s.
3. On continued failure: secondary deployment in the same region.
4. On failure of both: the complaint is queued for manual triage, the coordinator is alerted within 2 minutes (BRD NFR-10), and the system does not degrade to a lower-quality answer without saying so.

8.2 Guardrails

- Complaint text is wrapped in delimiters and passed as data, never concatenated into the instruction section of a prompt.
- Every agent has an explicit tool allow-list (Section 2.2). No agent holds a tool it does not need.
- No LLM-driven agent can call sap_order_create. That tool is held only by the deterministic supervisor node.
- Outputs are validated against a Pydantic schema before use. A response failing schema validation is retried once, then routed to a human — never coerced.
- System prompts are never echoed. A response containing more than 40 contiguous characters of the system prompt is rejected and logged as a suspected extraction attempt.

ON NFR-07 — PROMPT INJECTION

The complaint text is written by whoever emailed the service mailbox. Treat it exactly as you would treat a query string on a public web form.

The structural defences above matter more than any prompt wording. If your only mitigation is an instruction telling the model to ignore instructions, you do not have a mitigation.

9. Observability, Security and Evaluation

9.1 Trace fields

Every agent invocation emits a LangFuse trace carrying: trace_id, complaint_id, agent, model_id, prompt_version, input_token_count, output_token_count, latency_ms, confidence, retrieved_source_ids, tool_calls, outcome, and routing_reason where applicable. Traces to BRD FR-018 and NFR-08.

9.2 Metrics and alerts

Metric	Alert threshold	Channel	Traces to
Complaints ingested per hour	< 1 for 2 hours in business hours	Ops channel	NFR-03
End-to-end triage latency P95	> 90 seconds for 15 minutes	Ops channel	NFR-01
Auto-approval rate	Deviates more than 15 points from the 7-day mean	FDE team	FR-009
Supervisor override rate	> 30% over 24 hours	FDE team	FR-012, FR-020
Model cost per complaint	> ₹28 rolling 24-hour mean; and at 80% of monthly budget	FDE team + delivery manager	NFR-09
SAP write failure rate	> 2% over 1 hour	Ops channel + R. Balan	FR-017, INT-002
Citation-less diagnosis count	> 0	FDE team	FR-006

OVERRIDE RATE IS YOUR REAL ACCURACY METRIC

Offline evaluation tells you how the system performs on the data you had. Supervisor override rate tells you how it performs on the data you have.

It is also the metric that will move first when something breaks upstream — a changed email template, a new equipment model, a depot reorganisation. Watch it more closely than any benchmark.

9.3 Access control

Role	Can	Cannot
Coordinator	View all complaints, submit phone transcripts, view recommendations	Approve recommendations, change thresholds
Depot supervisor	All coordinator rights, plus approve, amend and reject for their own depot	Act on another depot's queue, change thresholds
Head of Customer Service	View across all depots, view all reports and audit trails	Approve individual recommendations
Administrator	Change confidence thresholds, ranking weights and escalation recipients	View complaint personal data
AtliQ engineer	View traces, metrics and pseudonymised records in production	View reporter personal data or approve recommendations

9.4 Evaluation

- Regression suite of 200 labelled historical complaints, run in CI on every change to a prompt, model version or rule table.
- Adversarial suite of 40 prompt injection attempts (BRD NFR-07). Zero successes permitted; a single success blocks release.
- Citation validity check: every probable cause must resolve to a retrievable manual page or service order. Failures block release.
- Shadow-mode agreement report produced daily throughout Wave 0, comparing AI recommendation to coordinator decision.

10. Traceability Matrix

Every functional requirement in BRD v1.1 maps to at least one section of this document. Non-functional requirements are addressed in Sections 7, 8 and 9.

BRD ref.	Requirement	Designed in TDD section
FR-001	Multi-channel intake	2.2, 4.1, 7 (INT-005, INT-006)
FR-002	Structured field extraction	5.1, 8
FR-003	Equipment identification	5.1, 5.2 (RT-05), 7 (INT-001)
FR-004	Duplicate detection	2.2, 4.1 (idempotency), 8
FR-005	Severity classification	6.1
FR-006	Probable fault diagnosis	2.2, 8, 9.4
FR-007	Skill code determination	5.1, 6.3
FR-008	Spare parts prediction	2.2, 5.1
FR-009	Confidence threshold and routing	5.2
FR-010	Technician matching	6.3
FR-011	Recommendation with rationale	2.2, 8
FR-012	Supervisor override	3 (Override entity), 5.3, 9.3
FR-013	Depot stock availability check	6.4, 7 (INT-003)
FR-014	Parts shortfall handling	6.4 (PRT-02 to PRT-04), 5.2 (RT-04)
FR-015	SLA tier resolution and clock start	6.2, 7 (INT-004), 4.1
FR-016	SLA breach prediction and escalation	6.2 (ESC-01 to ESC-04)
FR-017	SAP PM work order creation	7 (INT-002), 5.3
FR-018	Decision audit trail	3 (DecisionAudit), 9.1
FR-019	Supervisor console	4 (/v1/queue), 9.3
FR-020	Correction capture and reuse	3 (Override), 9.2

Commented [SFR7]: Do this last, and do it honestly. A traceability matrix is worthless if you fill it in by assuming rather than checking. Its real value is the gaps it exposes. This one was built at v0.9 and revealed that FR-004 had no design at all — it had been discussed and never written down.

HOW TO USE THIS IN AN INTERVIEW

If you are asked how you make sure nothing gets dropped between requirements and build, this table is the answer, and it is a much better answer than 'good communication'.

Build it in reverse as well: walk every section of your design and ask which requirement it serves. Anything that serves none is scope you invented, and it is costing someone money.

Annex A — Worked Sequence: Complaint to Work Order

A single P1 complaint traced end to end, showing which component acts, what it produces and which specification governs it. Use this to check your understanding of Sections 5 to 8.

#	Component	Action	Governed by
1	WhatsApp adapter	Receives 'Pump at Unit 3 has stopped completely, plant is down'. POSTs to /v1/complaints with source_message_id.	4.1
2	Orchestrator	Persists Complaint, starts the SLA clock at channel receipt time, state RECEIVED.	FR-015, 5.3
3	Intake agent	Extracts customer, site, symptom, reported time. reporter_phone present, customer_id resolved. State EXTRACTED.	5.1, 8
4	Intake agent	Checks for an open order on the same equipment within 72 hours. None found.	FR-004
5	Triage agent	Matches equipment to 000000418823 with confidence 0.94. Above the 0.80 threshold, so no ambiguity routing.	5.1, RT-05
6	Triage agent	Classifies fault_class = STOPPAGE with confidence 0.91.	5.1, 8
7	Rule engine	equipment.is_critical = true. SEV-R1 fires. severity = P1, severity_rule_fired = SEV-R1.	6.1
8	Triage agent	Reads AMC contract: tier GOLD. P1 response window 4 hours. sla_due_at computed.	6.2, INT-004
9	Triage agent	Retrieves manual and history; proposes three probable causes, each with a citation. Skill codes PM-ROT-02 and PM-SEAL-01 determined.	FR-006, FR-007
10	Parts agent	Predicts mechanical seal 40028811 as High likelihood. SAP MM shows 2 units at Pune depot. PRT-01 applies.	6.4, INT-003
11	Dispatch agent	Four technicians survive the hard filters. Scores computed. Top candidate 0.87. Rationale generated.	6.3
12	Supervisor node	severity = P1, so RT-01 fires unconditionally. State PENDING_REVIEW, routing_reason P1_MANDATORY_REVIEW.	5.2
13	Depot supervisor	Opens the queue, sees the item at the top (lowest SLA time remaining), reads the rationale, approves without amendment.	FR-019, 9.3
14	Supervisor node	Calls sap_order_create through CPI with idempotency key rec_id. Order 4000318842 returned. State ORDER_CREATED.	INT-002
15	Notification	WhatsApp template message sent to the assigned technician.	INT-006
16	Audit	DecisionAudit rows written for every agent invocation with model ID, prompt version and confidence.	9.1, FR-018
17	Scheduler	ESC-01 armed at 80% of the 4-hour window; ESC02 at 100%.	6.2

Total elapsed time from step 1 to step 12: 41 seconds in the Wave 0 measurement. Steps 13 onward depend on supervisor response time, which is outside the NFR-01 latency budget and is measured separately.

Annex B — The Spec Clarity Test

There is one honest way to find out whether a technical design is finished: hand it to someone who was not in the room and see whether they build the right thing. An AI coding agent is an excellent stand-in, because it will not fill your gaps with charity — it will fill them with plausible invention, which is exactly what a rushed developer does.

The exercise

5. Open a coding agent (Claude Code, Cursor, or similar) in an empty repository.
6. Paste Section 6.1 (severity decision table) and Section 6.3 (technician ranking) with no other context, and ask for a Python module implementing both, plus unit tests.
7. Do not answer any clarifying question the agent asks. Write each question down instead.
8. Run the tests. Then read the code against the specification, line by line.

What you are looking for

Signal	What it means
The agent asked no clarifying questions and the code matches	The specification is complete for that section. This is the target.
The agent asked a question you cannot answer from the document	You found a genuine gap. Every question on your list is a defect in the spec, not in the agent.
The code runs but implements a subtly different rule	The most dangerous outcome. Something in your wording admitted a second reading. Find it and remove the ambiguity.
The agent invented a rule that is not in the specification	You left a decision implicit and something filled it in. A developer under deadline pressure does the same thing, only slower and less visibly.

Then try it on a weaker spec

Repeat the exercise with the rejected wording from Annex B of the BRD — 'priority will be assigned appropriately based on the situation' — and compare what comes back. The gap between the two runs is the entire value of a technical design document, demonstrated in about ten minutes.

WHY THIS MATTERS MORE EVERY YEAR

As more implementation is done by coding agents, the specification stops being paperwork on the way to the real work and becomes the input to the real work.

The FDE skill that appreciates fastest is not writing code. It is writing a specification precise enough that the code is a formality — and being able to tell, by reading it, whether what came back is right.

Annex C — TDD Reviewer's Rubric

Grade your own technical design before handing it to anyone. Any No in the Blocking column means it is not ready for build.

#	Check	Blocking	Where here
1	Does every functional requirement in the BRD map to at least one design section?	Yes	Section 10
2	Is every business rule expressed as a table or formula rather than prose?	Yes	6.1, 6.2, 6.3, 6.4
3	Are rules ordered, with first-match-wins stated explicitly and the matched rule ID persisted?	Yes	6.1
4	Does every default fail towards attention rather than away from it?	Yes	SEV-R9, INT-003, INT-004
5	Does every validated field say what happens on failure, and does it forbid guessing?	Yes	5.1
6	Does every integration state its failure behaviour, retry policy and idempotency key?	Yes	Section 7, 4.1
7	Is there a defined route to a human, and is it impossible for the model to bypass it?	Yes	5.2, 2.2
8	Can any LLM-driven component write to the system of record?	Must be No	2.2, 8.2
9	Are model IDs and prompt versions pinned and recorded on every decision?	Yes	Section 8, 9.1
10	Is there a cost design, not just a cost target — i.e. is the premium model justified per task?	Yes	Section 8
11	Are prompt-injection defences structural, not just instructional?	Yes	8.2
12	Is every error code enumerated with the client behaviour it implies?	Yes	4.1
13	Is the state machine complete, with invalid transitions rejected rather than undefined?	Yes	5.3
14	Could someone who has never spoken to the client build Section 6 without asking a question?	Yes	Annex B
15	Are configurable values marked as configuration, with who can change them and whether a redeploy is needed?	No	5.2, 6.3
16	Is there a worked end-to-end example a new joiner can follow?	No	Annex A

CHECK 8 IS THE ONE TO REMEMBER

It is the only row in this rubric whose correct answer is No. If any language model in your architecture can write directly to the client's system of record, you have built something that cannot be secured against its own inputs.

Put a deterministic gate in front of every irreversible action. That is the design pattern the rest of this document is built around, and it is the answer to the question every enterprise security review will eventually ask you.